

placed between the probes, assume that S1 is turned to ground resistor R5 (10-kilo-ohm). If the resistance between the probes is close to half the supply level, a 470Hz note is generated, which is easily distinguishable from 910 Hz. When the resistance between the probes is greater than 10 kilo-ohms, the pitch is lower. As the resistance approaches 100 kilo-ohms, the pitch falls to the buzzing level. If S1 is switched now to bring R6 (100-kilo-ohm) into the circuit, the input rises again to half the supply level and the pitch rises to 470 Hz.

Summing up, if S1 is switched to a given range and a medium-pitched note is heard, the resistance between the probes is of the same order as the selected resistor. If a buzz is heard, switch to a higher range and vice-versa.

Switching range is necessary only if you want a rough measure of the resistance. If you are interested only in

finding a short-circuit or open circuit, you need not bother about setting the range. The output of IC1 at pin 4 is fed to transistor T1, causing it to rapidly turn on and turn off through resistor R3. The current pulsing through the loudspeaker produces the buzzing sound or tones.

This circuit can also be used for checking diodes and transistors. If the anode of the diode under test is connected to positive probe 1 and the cathode to probe 2, the diode becomes forward-biased. Current flows through it and the input voltage to the IC rises producing a medium-pitched note. When the diode is the other way round, no current flows through it. The input is pulled down to 0V and a low buzzing sound is heard.

You can check bipolar junction transistors (normal transistors) in the same way, treating the junctions as diodes. For the base-emitter junc-

tion of an npn transistor, the base is anode and the emitter is cathode. For the base-collector junction, the base is anode and the collector is cathode. The pnp transistors are much the same; only the anodes and cathodes are reversed. As a quick test, checking either type of transistors with the probes to the collector and emitter should give a buzzing sound.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. S1 is a single-pole, four-way switch. In place of S1, a four-way, dual in-line miniature switch can also be used. Connect its four pins along one side to the ground and the remaining four pins on the other side to the resistors. Mount switch S1 on the front side of the case with its common terminal connected to the ground. Use a pair of mini test clips, which are generally easier to attach to the pins of ICs. Banana plugs can also be used. ●

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